



SAINT PIERRE
LEARNING RESOURCES

SCIENCE

Elements & Compounds

Particle Evidence Lab

Grade 7 • Texas

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Teacher Guide

Level: Grade 7 • Texas

Standard / objective: TEKS 7.6(A) — Compare and contrast elements and compounds in terms of atoms and molecules, chemical symbols, and chemical formulas.

Source / course note: Texas Education Agency TEKS Guide - Science 7.6A:
<https://teksguide.org/teks/s76a/overview>

Before this lesson: Recognize an atom and read simple element symbols.

Model and discuss: Compare O_2 and CO_2 : O_2 has two atoms of one element; CO_2 has three atoms of two elements. Count kinds of atoms before classifying. $NaCl$ represents a ratio in an ionic compound, not a discrete molecule.

Watch for: More than one atom does not automatically mean compound. The number of different elements matters.

Suggested sequence: 5-minute launch, 10-minute model, 20-30 minutes of practice, then a 5-minute exit ticket. Use separate paper for extended writing.

Learning supports

AIG: Explain why both O_2 and O_3 are elements even though their molecules differ.

504: Offer chunked tasks, reduced copying, extra time, and breaks as specified in the learner's plan.

EC: Use labeled counters for atom types; keep labels as well as colors.

ESL: Rehearse: This is a ___ because it contains ___ kind(s) of atom.

Adapt these suggestions to the current learning plan. Print on US Letter paper at actual size; keep the key separate. One teacher may use the resource with their own students; no public posting or resale.



Student Directions & Practice

Analyze particle models and formulas to distinguish elements from compounds.

Directions: Complete every item. Show work or cite evidence. For open responses, answer in complete sentences.

1. Classify O_2 as element or compound and explain.
2. Classify H_2O and name the kinds and number of atoms.
3. What does the 2 mean in CO_2 ?
4. Compare Na and NaCl.
5. A model has four identical N atoms. Element or compound?
6. A model has two molecules, each with two H and one O. Write the formula.
7. Explain why air is a mixture, not a compound.
8. Design a particle key for CO_2 and sketch three molecules.



Assessment / Exit Ticket

Work independently. Show your reasoning. Use extra paper if needed.

1. Classify N_2 and CO_2 as element or compound. Explain each.
2. Consider two separate water molecules (H_2O). How many H and O atoms are present in total?

Reflection: What helped you decide? What do you want to practice next?

Score each task: 2 = accurate with clear reasoning; 1 = partly correct or missing reasoning; 0 = incorrect or no evidence. The reflection is not scored.

Answer Key & Teaching Guidance

1. Element; it contains only oxygen atoms.
2. Compound; two hydrogen atoms and one oxygen atom.
3. Two oxygen atoms in each molecule.
4. Na is an element symbol; NaCl is a compound formula containing sodium and chlorine.
5. Element.
6. H_2O .
7. Its substances are physically combined in variable proportions.
8. Three groups, each with one carbon and two oxygen particles.

Exit ticket guidance

1. N_2 is an element (nitrogen only). CO_2 is a compound (carbon and oxygen chemically combined).
2. Four hydrogen atoms and two oxygen atoms. Each H_2O molecule contains two H and one O.

Next teaching move: Use the worked model again if the concept is unclear. If the answer is correct but reasoning is incomplete, ask for an explanation. Offer the extension once both tasks are secure. Scores guide instruction; they are not standardized mastery cutoffs.